

Vijay Aleti

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PROFESSIONAL SUMMARY:

- Data Engineer with **5+ years** of hands-on experience designing, testing, and maintaining enterprise-scale data pipelines for advanced analytics.
- Proven expertise in **SQL, data warehousing, ETL/ELT tools, and data transformation** with demonstrated success in cleansing complex datasets and optimizing performance across production environments.
- Skilled in **Python object-oriented programming** and cloud-native architecture. Seeking to leverage **DataStage maintenance experience** while driving transition to **IBM Cloud Pak** for scalable, AI-ready data solutions.
- 5+ years SQL development and data engineering expertise across enterprise platforms
- Proficient in ETL tools including **Informatica, Azure Data Factory, and Databricks** with transferable DataStage knowledge
- 2+ years hands-on **Snowflake** experience with query optimization and data warehousing
- Advanced **Python OO programming** (Pandas, PySpark, NumPy, SQLAlchemy) and SQL (T-SQL, PL/SQL)
- 2+ years cloud platform experience (**Azure** - ADLS Gen 2, Synapse, ADF; **AWS** - S3, Redshift)
- Proven ability to design scalable data models with performance tuning and optimization

TECHNICAL SKILLS

ETL/ELT & Data Engineering: Informatica PowerCenter, Azure Data Factory, Databricks/PySpark, Apache Spark, dbt, Airflow, Kafka, Big Data technologies, Change Data Capture (CDC), Delta Lake, Schema validation

Programming & Scripting: Python (Pandas, PySpark, NumPy, SQLAlchemy), SQL (T-SQL, PL/SQL, Advanced Query Optimization), Bash, YAML, JSON

Data Warehousing & Databases: Snowflake (clustering, partitioning, cost optimization), Azure Synapse Analytics, Google BigQuery, Azure SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, Redis, Azure Cosmos DB

Cloud Platforms: **Azure:** ADLS Gen 2, Data Factory, Synapse, Databricks, Functions, App Service, HDInsight, Monitor

AWS: S3, Redshift, Lambda, Kinesis

Data Model & Performance Optimization: Slowly Changing Dimensions (SCDs), Partitioning strategies, Materialized views, Indexing, Query optimization, Cost tuning

DataOps & Infrastructure: Docker, Kubernetes, Terraform, Ansible, Azure DevOps, GitLab CI/CD

Data Governance & Security: Data lineage, Metadata management, Master Data Management (MDM), Azure Purview, GDPR, HashiCorp Vault, IAM, RBAC

Monitoring & Observability: Azure Monitor, Application Insights, Grafana, Prometheus, Splunk, Datadog

Business Intelligence: Power BI, Tableau, Looker, Qlik Sense

PROFESSIONAL EXPERIENCE

Optum

Analytics Engineer | Remote | Sep 2025 - Present

- **Built automated enterprise data pipelines** ingesting 5M+ records daily from Azure Data Lake to **Snowflake** using Azure Data Factory, ensuring 99.9% pipeline uptime and compliance with information governance best practices

- **Optimized Snowflake query performance** through intelligent clustering and partitioning strategies, achieving **35% improvement** in analytical response times and reducing warehouse costs by **20%**
- **Advanced ML/AI integration:** Enabled real-time model retraining by integrating **Snowflake datasets with Azure Machine Learning**, reducing experiment setup time by 50% and supporting data pipeline automation
- **Maintained robust observability** through Azure Monitor proactive monitoring and automated alerting, supporting secure, reliable data operations at enterprise scale

Clairvoyant

Data Engineer Intern | Remote | **Jan 2025 - May 2025**

- **Designed scalable ETL workflows** in Azure Data Factory, PySpark, and Snowflake, transforming and cleansing **10M+ daily records** with **40% improvement** in downstream analytics accuracy
- **Engineered data models for performance:** Optimized Azure Synapse table configurations and applied compression techniques, lowering compute expenses by **20%** while strengthening query efficiency
- **Advanced analytics pipeline:** Streamlined distributed analytics using Azure HDInsight, Azure Synapse Serverless SQL, and ADLS Gen 2, improving query speed by **35%** for near real-time insights
- **Automated data quality:** Developed Python monitoring utilities integrated with Azure Monitor and Event Grid, proactively detecting pipeline bottlenecks and reducing mean resolution time by **30%**
- **Collaborated on unified data lakehouse** integrating structured and semi-structured sources using Azure Data Lake Storage, Azure Purview, and Azure Synapse, improving governance compliance

Accenture

Data Engineer | Remote | **Jun 2021 - Sep 2023**

- **Designed & optimized enterprise-scale ETL pipelines** on **Azure Data Factory** and **SQL Server**, handling structured/unstructured datasets with improved query performance and reliability
- **High-performance distributed processing:** Leveraged **Databricks with PySpark** to transform **20M+ daily records**, improving pipeline efficiency by **30%** and accelerating business insights
- **Advanced data model optimization:** Enhanced analytical performance in Azure Synapse through intelligent partitioning, materialized views, and indexing, achieving **35% reduction in query runtimes** and \$50K annual cost savings
- **Implemented CDC logic** within ADF and Delta Lake, shrinking data refresh cycles from 24 hours to **<2 hours**, enabling near real-time visibility for enterprise dashboards
- **Data quality & governance:** Embedded validation checks into ADF Mapping Dataflows and SQL procedures, decreasing transformation errors by **25%** while enforcing GDPR and audit compliance
- **Automated ML/AI data pipelines** in support of machine learning initiatives, preparing and maintaining datasets for advanced analytics
- **Designed lakehouse architecture** unifying structured (Synapse, SQL) and semi-structured (JSON, Parquet) sources, expanding self-service analytics to 500+ business users

Sonata Software

Data Engineer | Remote | **Jun 2019 - May 2021**

- **Delivered 10+ scalable ETL pipelines** using **Azure Data Factory** and **Azure Functions**, improving ingestion of structured/unstructured datasets and cutting manual processing by **25%**
- **Designed enterprise ETL workflows** with **Informatica PowerCenter**, creating standardized mappings and configurations that simplified complex jobs and improved cross-team reusability
- **Implemented event-driven architecture** using Azure Logic Apps and Event Grid, replacing static scheduling and reducing end-to-end processing time by **30%**

- **Data model optimization & SCD handling:** Applied slowly changing dimension patterns and partitioning strategies, optimizing warehouse performance for real-world applications
- **Migrated 5TB+ enterprise data** from Oracle and Teradata into Azure Data Lake and Azure Synapse using Azure Database Migration Service, achieving **40% faster query performance**
- **Advanced Spark tuning & streaming:** Optimized Databricks Spark jobs for streaming pipelines, increasing throughput by **25%** and improving reliability through error-handling and schema validation logic
- **Enabled 200+ business users** with Lakehouse setup on Databricks, Azure Synapse, and Azure Data Lake, directly supporting self-service analytics adoption

EDUCATION

Master of Science, Information Systems

Saint Louis University | Aug 2023 - May 2025

Bachelor of Technology, Mining Engineering

Jawaharlal Nehru Technological University | Jul 2016 - May 2019

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **SnowPro Advanced Data Engineer Certification** — Snowflake
- **IBM Data Engineering Professional Certificate** — Coursera • **Azure Fundamentals (AZ-900)** — Microsoft
- **Security, Compliance, and Identity Fundamentals (SC-900)** — Microsoft
- **Data Engineering with Databricks** — DataCamp
- **Oracle Certified AI Generative Professional**

PROJECTS & ACHIEVEMENTS

Physician Burnout Analysis Using ML & Dash

- Built end-to-end data ingestion and transformation pipelines using Python, Pandas, and SQL, consolidating workload data from multiple sources into structured datasets
- Applied advanced analytics (K-Means clustering with Scikit-learn) to identify actionable workforce insights
- Deployed analytics platform on Azure App Service with RBAC-based access and real-time dashboarding capabilities